

**GATE-ALL BRANCHES
ENGINEERING MATHEMATICS**



SINGLE VARIABLE CALCULUS

Lecture No.- 12



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Recap of previous lecture



Topic

Concepts based on definite integrals

Topic

Area based problems

Topic

Improper integral

Topics to be covered



Topic

Concept of taylor's series

Topic

Problems based on taylors series

Topic

Partial derivatives

Topic

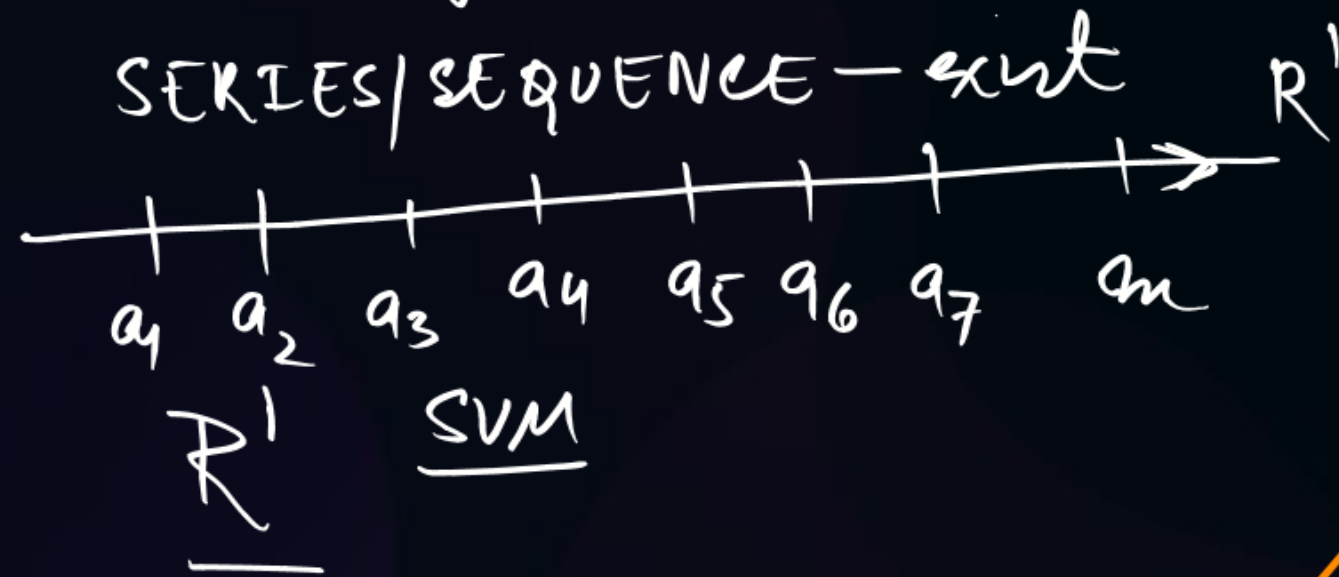
Problems based on partial derivatives

✓ Imp. Taylor SERIES :

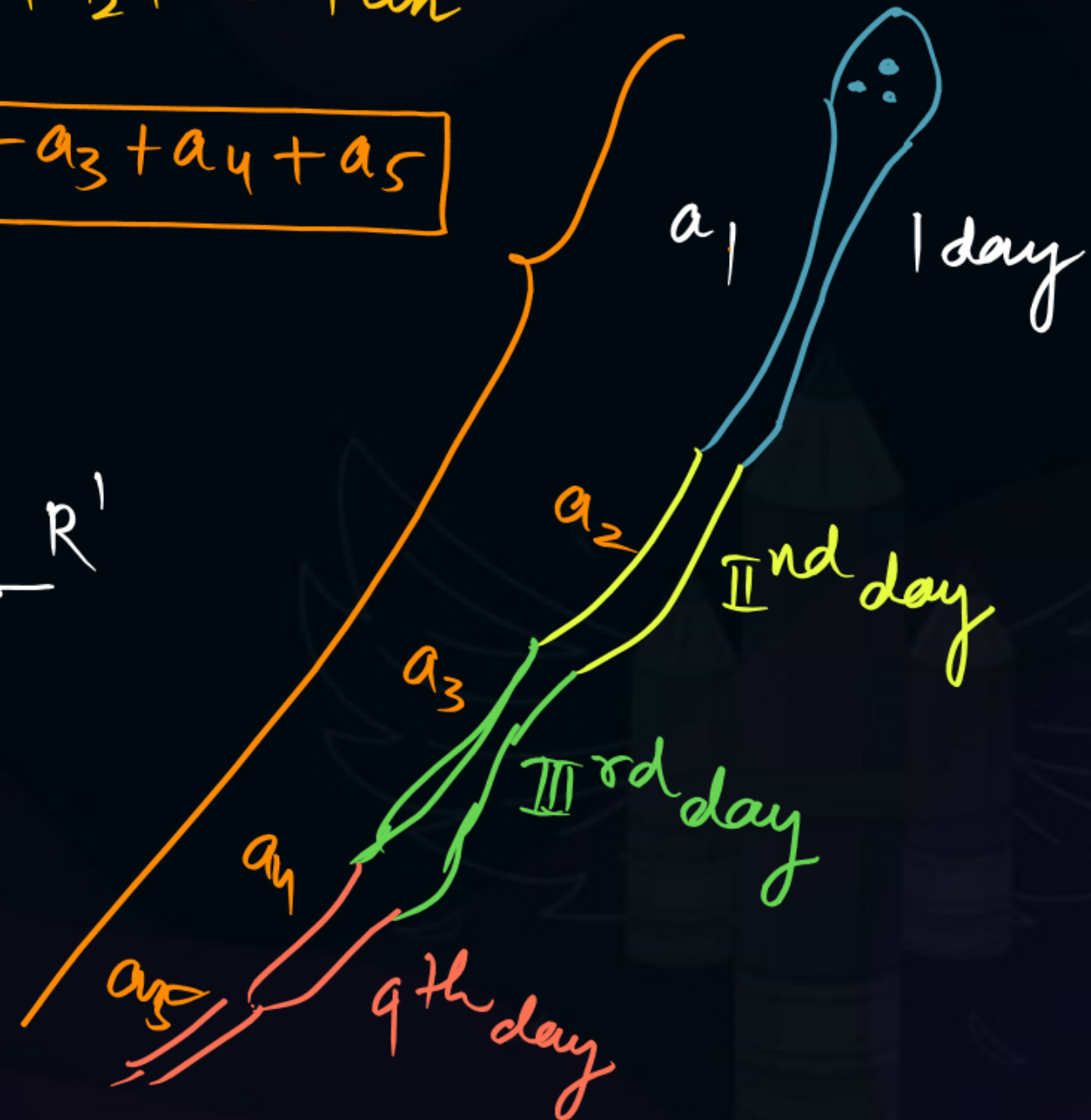
Sequence : $a_n = a_1 + a_2 + \dots + a_n$

← $a_5 = a_1 + a_2 + a_3 + a_4 + a_5$
5 day growth

↙ convergence



Snake growth



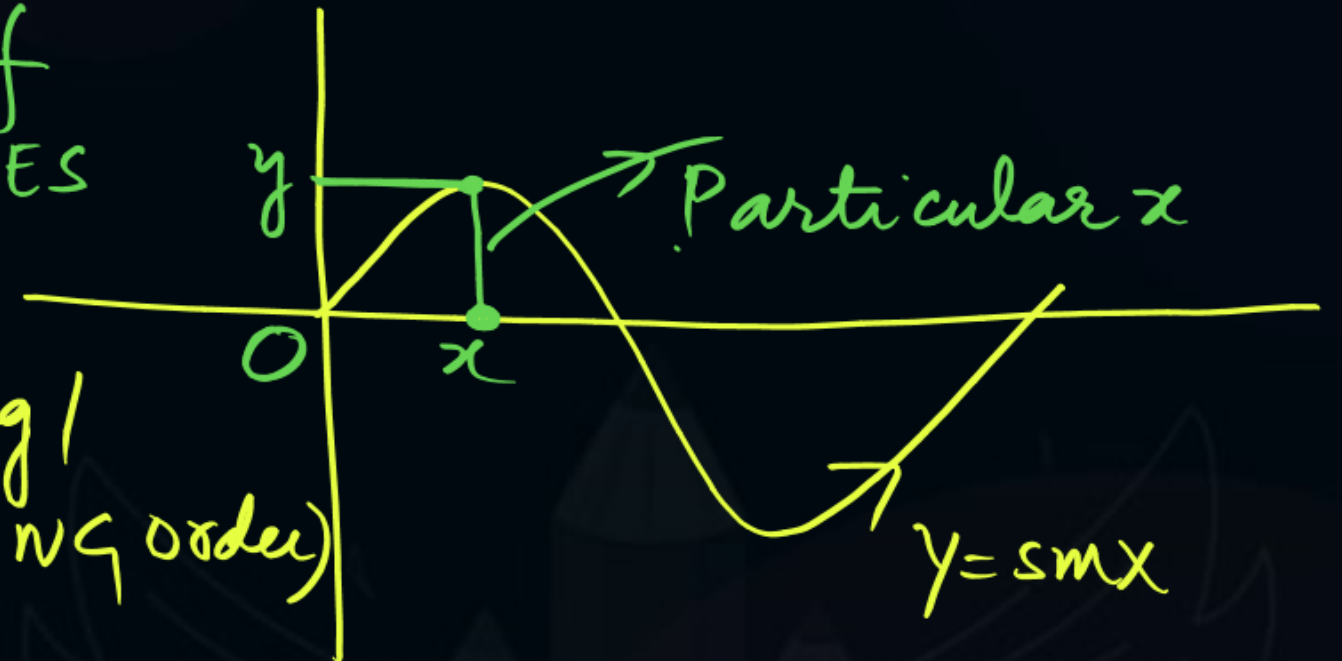
SERIES:

Ascending order of x

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_nx^n$$

Any Function represent Term of
 $\sin x, \cos x, \tan x, e^x, \log x,$ SERIES

Algebraic Function (Ascending /
 DESCENDING order)

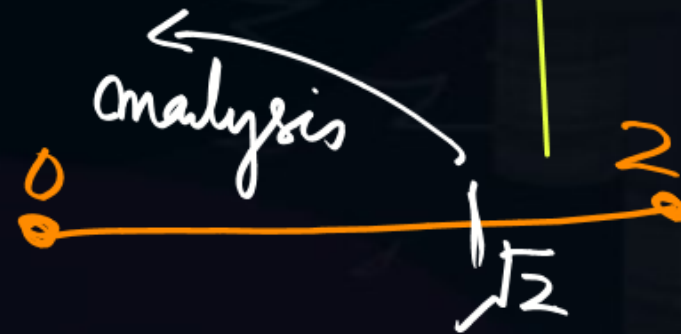
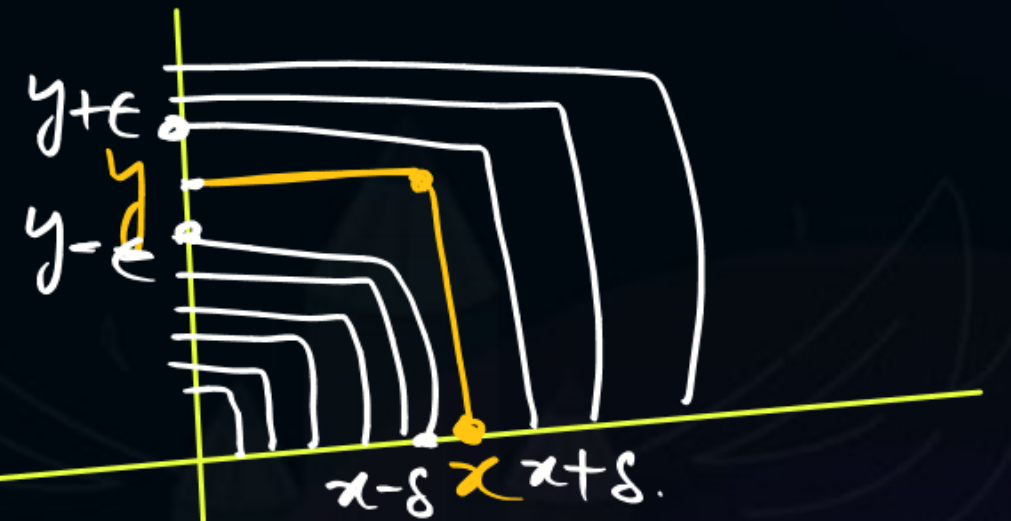
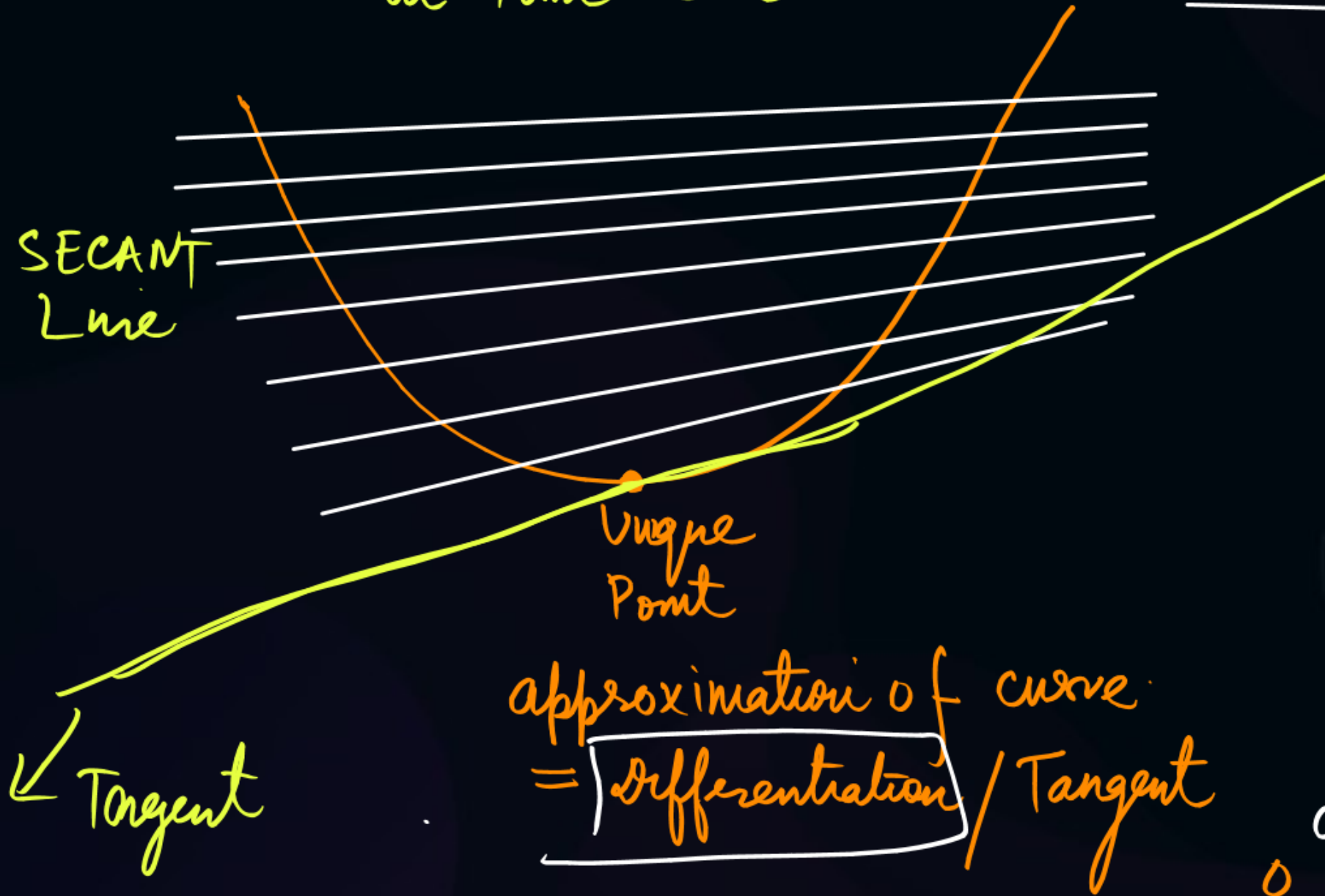
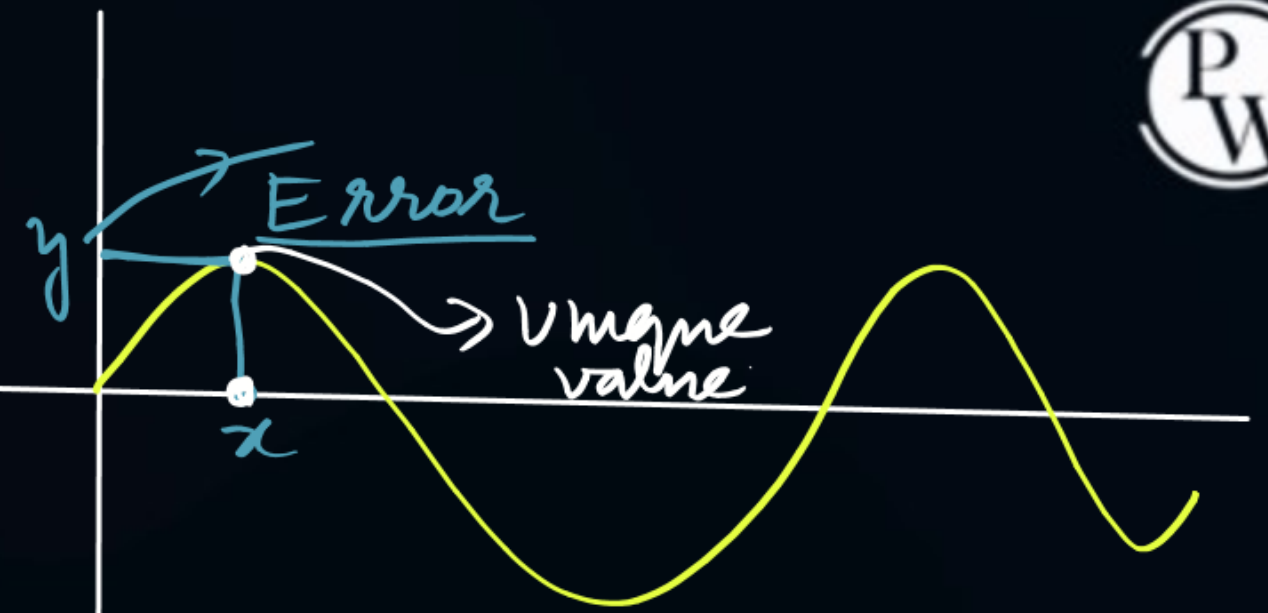


$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_nx^n$$

$a_0, a_1, a_2, \dots, a_n$ Are constants

$$f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$$

at Point $x=0$



$a_0, a_1, a_2, \dots, a_n$ are constant

$$\checkmark f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n \quad \text{Put } x=0 \quad \boxed{f(0) = a_0}$$

Differentiation w.r.t x

$$f'(x) = 0 + a_1 + 2a_2x + 3a_3x^2 + 4x^3a_4 + \dots + a_nx^{n-1}$$

Put $x=0 \quad \boxed{f'(0) = a_1}$

Again Approximation

$$f''(x) = 2a_2 + 6a_3x + 12x^2a_4 + \dots$$

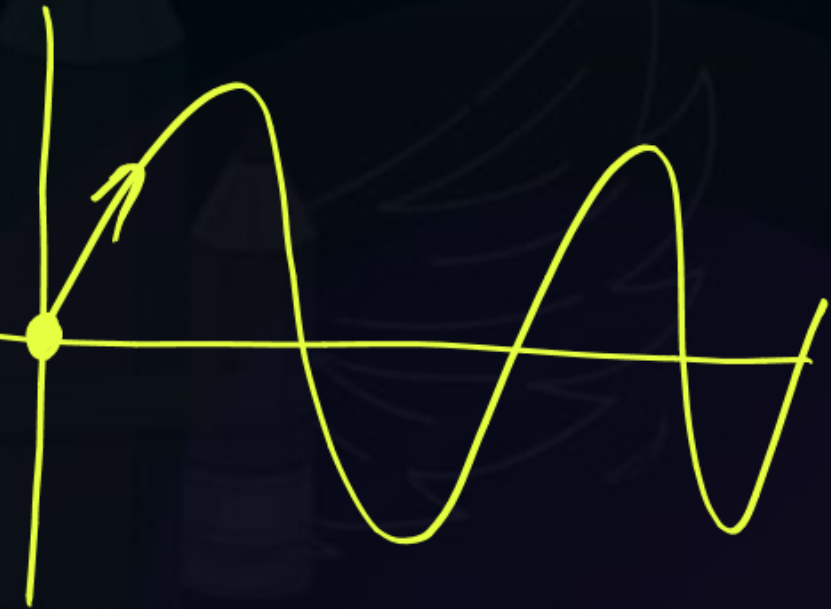
$x=0 \quad \boxed{\frac{f''(0)}{2!} = a_2}$

$$f'''(x) = 6a_3 + 24xa_4 + \dots$$

$x=0 \quad \boxed{\frac{f'''(0)}{3!} = a_3}$

$$f^{(4)}(x) = 24a_4 + \dots$$

$x=0 \quad \boxed{\frac{f^{(4)}(0)}{4!} = a_4}$



graph

$$1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots$$

Put The value $a_0, a_1, a_2 \dots a_n$

$$f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots + a_n x^n$$

Maclaurin
SERIES

$$f(x) = f(0) + \frac{f'(0)}{1!} x + \frac{f''(0)}{2!} x^2 + \frac{f'''(0)}{3!} x^3 + \frac{f^{(4)}(0)}{4!} x^4 + \dots$$

$$f(x) = e^x$$

$$f(0) = e^0 = 1$$

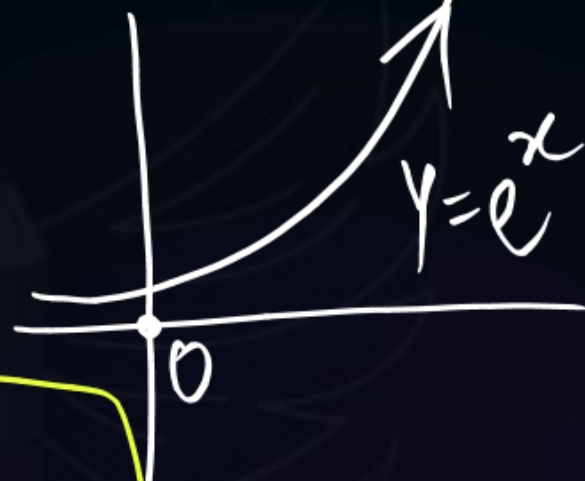
$$f'(0) = e^0 = 1$$

$$f''(0) = e^0 = 1$$

$$f(x) = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!}$$

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

at Pt $x=0$



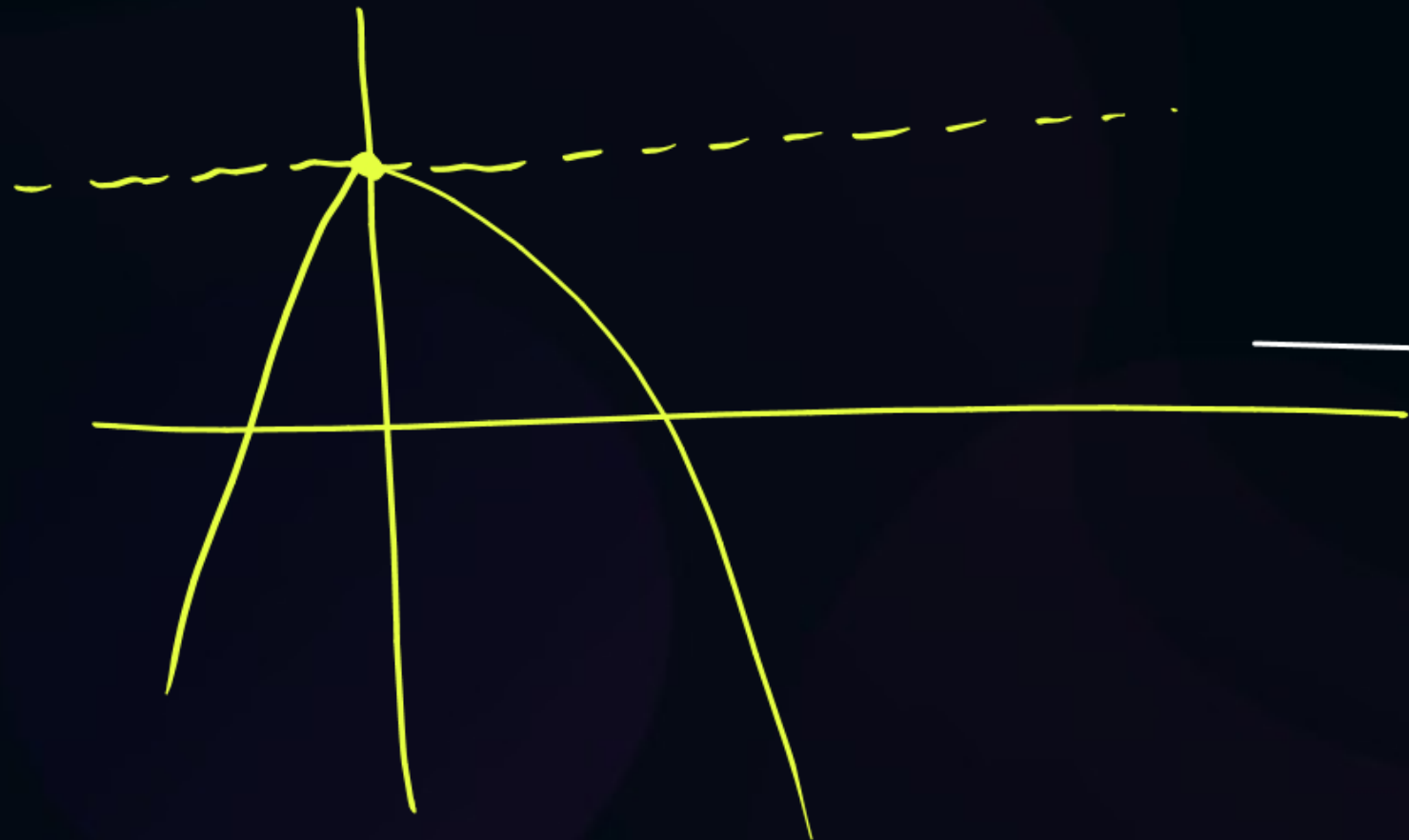
$$f(x) = \cos x$$

$$f(0) = \cos 0 = 1$$

$$f'(0) = -\sin 0 = 0$$

$$f''(0) = -\cos 0 = -1$$

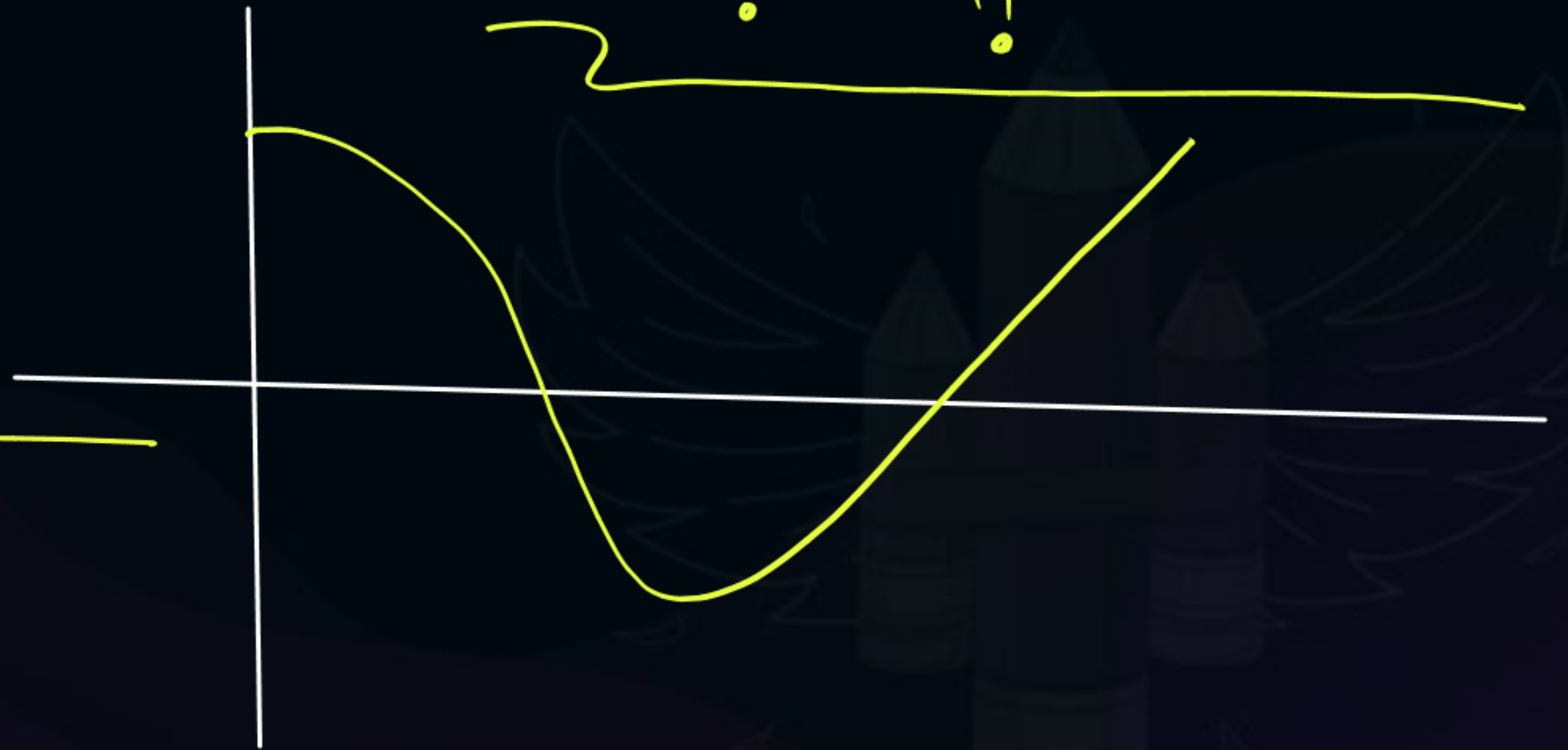
$$f'''(0) = \sin 0 = 0$$



$$f(x) = \cos x \text{ at Point } x = 0$$

$$f(x) = \cos x = f(0) + \frac{f'(0)x}{1!} + \frac{f''(0)x^2}{2!} + \dots$$

$$= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$



Using machine SERIES at $x=0$

$$f(x) = \sin x, e^{-x}, \boxed{\tan x}, \boxed{\log x}$$

more Term $\rightarrow$ more Precision



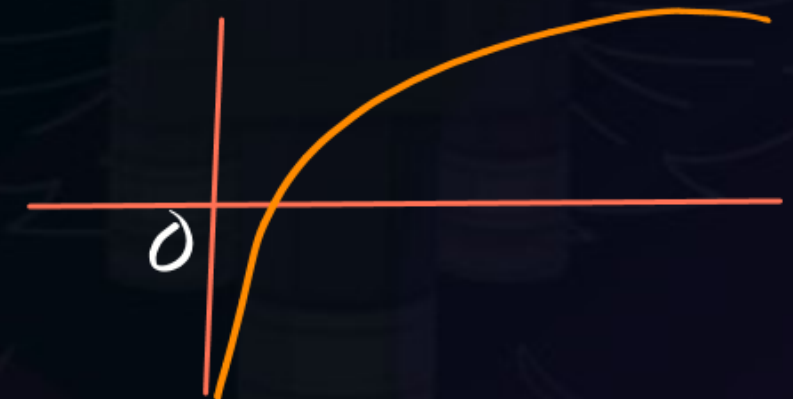
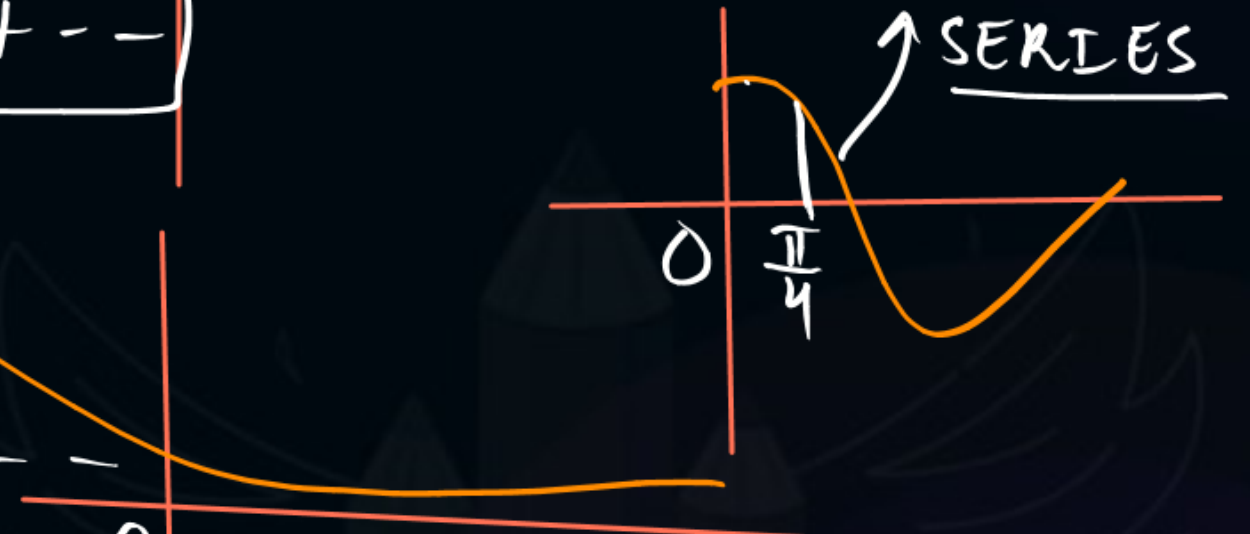
at Point $x=a$

$$f(x) = a_0 + (x-a)a_1 + a_2(x-a)^2 + a_3(x-a)^3 + \dots$$

at a Point $x=a$

$$f(x) = f(a) + \underbrace{(x-a)}_1 f'(a) + \underbrace{(x-a)^2}_2 f''(a) + \dots$$

Taylor SERIES at Point $x=a$



$$f(x) = f(a) + \underbrace{(x-a) \frac{f'(a)}{1!}}_{\text{Linear approximation}} + \frac{(x-a)^2}{2!} f''(a) + \frac{(x-a)^3}{3!} f'''(a) + \dots$$

Taylor SERIES
at $x=a$

$$\text{Linear approximation} = f(a) + (x-a) \frac{f'(a)}{1!} = \text{Differentiation}$$

$$f(x) = f(a) + (x-a) f'(a)$$

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{(x-a)} = f'(a)$$

Newton
first
derivative

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$



Questions



#Q. The third term in the Taylor's series expansion of e^x about 'a' would be ____.

(a) $e^a (x - a)$

(b) $\frac{e^a}{2} (x - a)^2$

(c) $\frac{e^a}{2}$

(d) $\frac{e^a}{6} (x - a)^3$

$$x = a$$

e^x Taylor SERIES

$$f(x) = f(a) + \underbrace{(x-a)}_{\text{I}} \underbrace{f'(a)}_{\text{II}} + \underbrace{\frac{(x-a)^2}{2!}}_{\text{III}} f''(a) + \dots$$

$$\begin{aligned} \text{Third term} &= \frac{(x-a)^2}{2!} f''(a) \\ &= \frac{(x-a)^2}{2!} e^a \\ &= \frac{e^a}{2} (x-a)^2 \end{aligned}$$



Questions



#Q. The Taylor series expansion of $\sin x$ about $x = \frac{\pi}{6}$ is given by

(a) $\frac{1}{2} + \frac{\sqrt{3}}{2}\left(x - \frac{\pi}{6}\right) - \frac{1}{4}\left(x - \frac{\pi}{6}\right)^2 - \frac{\sqrt{3}}{12}\left(x - \frac{\pi}{6}\right)^3 + \dots$

(b) $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$

(c) $\frac{x - \frac{\pi}{6}}{1!} - \frac{\left(x - \frac{\pi}{6}\right)^3}{3!} + \frac{\left(x - \frac{\pi}{6}\right)^5}{5!} - \frac{\left(x - \frac{\pi}{6}\right)^7}{7!}$

(d) $\frac{1}{2}$

$$= \sin \frac{\pi}{6} + \underbrace{\left(x - \frac{\pi}{6}\right)}_{1!} \cos \frac{\pi}{6} + \frac{\left(x - \frac{\pi}{6}\right)^2}{2!} \left(-\sin \frac{\pi}{6}\right) + \dots$$



Questions



$$x=a \checkmark$$

#Q. For the function e^{-x} , the linear approximation around $x=2$ is

$$\begin{aligned} \text{Linear app} &= f(a) + \underbrace{(x-a)}_{!!} \underbrace{f'(a)}_{!!} \\ &= e^{-2} + (x-2)(-e^{-2}) \\ &= \underline{(3-x)e^{-2}} \end{aligned}$$

(a) $(3-x)e^{-2}$

(b) $1-x$

(c) $\left[3+2\sqrt{2}-(1+\sqrt{2})x\right]e^{-2}$

(d) e^{-2}



Questions



#Q. In the Taylor series expansion of $(e^x + \sin x)$ about the point $x = \pi$, the coefficient of $(x - \pi)^2$ is

- (a) e^π
- (b) $0.5e^\pi$
- (c) $e^\pi + 1$
- (d) $e^\pi - 1$

$$f(x) = e^x + \sin x \text{ about } x = \pi$$

$$= (x - \pi)^2 \frac{f''(\pi)}{2!}$$

$$\frac{f''(\pi)}{2!} = \frac{e^\pi - \sin \pi}{2!}$$

$$= \frac{e^\pi - 0}{2!} = \frac{e^\pi}{2} = 0.5e^\pi$$

$$f'(\pi) = e^\pi + \cos \pi$$
$$f''(\pi) = e^\pi - \sin \pi$$

$$f(a) + (x-a) \frac{f'(a)}{1!} + (x-a)^2 \frac{f''(a)}{2!} + \dots$$

← 5(x)
coefficient



Questions



#Q. The Taylor series expansion of $\frac{\sin x}{x-\pi}$ at $x = \pi$ is given by

(a) $1 + \frac{(x-\pi)^2}{3!} + \dots$

(b) $-1 - \frac{(x-\pi)^2}{3!} + \dots$

(c) $1 - \frac{(x-\pi)^2}{3!} + \dots$

(d) $-1 + \frac{(x-\pi)^2}{3!} + \dots$

✓ Differ Not valid
means.

✓ Taylor SERIES
Not applicable

Maclaurins SERIES

$$f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$$

$x=0$

$$f(x) = f(a) + \frac{(x-a)}{1!} f'(a) + \frac{(x-a)^2}{2!} f''(a) + \dots$$

$x=a$
Shift

NECESSARY
and sufficient
condition

→ both are exists
only differentiable Function

$$f(x) = \frac{\sin x}{(x-\pi)} \quad \text{at } x=\pi$$

$$f(\pi) = \frac{\sin \pi}{\pi - \pi} = \text{undefined}$$

→ Diff - Not exists

Put $x - \pi = t$
 $x = (\pi + t)$

Collen

$$f(t) = \frac{\sin(\pi + t)}{t} = -\frac{\sin t}{t} \quad \text{at point } t=0$$

$\sin t$
 expand $t=0$
 Using maclaurins
 SERIES

$$= -\frac{1}{t} \left[t - \frac{t^3}{3!} + \frac{t^5}{5!} - \dots \right]$$

Using maclaurins SERIES

$$= -1 + \frac{t^2}{3!} - \frac{t^4}{5!} + \dots$$

$$= -1 + \frac{(x-\pi)^2}{3!} - \frac{(x-\pi)^4}{5!} + \dots$$

Ans



Questions



#Q. The infinite series $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$ corresponds to

gate

- (a) $\sec x$
- (b) e^x
- (c) $\cos x$
- (d) $1 + \sin^2 x$



Questions



H.W

#Q. The Taylor series expansion of $(3 \sin x + 2 \cos x)$ is

(a) $2 + 3x - x^2 - \frac{x^3}{2} + \dots$

(b) $2 - 3x + x^2 - \frac{x^3}{2} + \dots$

(c) $2 + 3x + x^2 + \frac{x^3}{2} + \dots$

(d) $2 - 3x - x^2 + \frac{x^3}{2} + \dots$



Questions



#Q. Let $f(x) = e^{x+x^2}$ for real x . From among the following, choose the Taylor series approximation of $f(x)$ around $x = 0$, which includes all powers of x less than or equal to 3.

(a) $1 + x + x^2 + x^3$

(b) $1 + x + \frac{3}{2}x^2 + x^3$

(c) $1 + x + \frac{3}{2}x^2 + \frac{7}{6}x^3$

(d) $1 + x + 3x^2 + 7x^3$

$e^{x+x^2} = f(x)$ Less Power 3

H.W

THANK - YOU